

NOTES FROM

80/20 Running

Matt Fitzgerald

Running isn't always supposed to be hard. In fact, most of the time, it should be easy and enjoyable.

A recovery run serves to increase running volume in a gentle way that does not interfere with recovery from recent hard training.

bike training had a strong positive effect on running performance, but swimming did not.

Too much hard running is the most common mistake in the sport.

In other sports, including running, the 80/20 Rule applies on a time basis. Top runners everywhere spend close to eight out of every ten minutes of training time at low intensity

Tolerance for suffering is also trainable.

Generally, experienced runners and fitter runners have the most variable strides, beginners the least

moderate intensity is represented by perceived effort ratings of 5 to 6, low intensity by ratings of 1 to 4, and high intensity by ratings of 7 to 10.

pace is the best choice for use as the primary gauge of intensity in high-intensity workouts.

Outside of the elite ranks, few runners discover their personal maximum running volume. Running just isn't important enough for most recreational runners to make that kind of commitment. Each runner must decide how much time and effort he's willing to commit toward improvement.

I recommend that you increase the total number of workouts in your typical training week by no more than one or two sessions at a time

Endurance was the true limitation in running

training for increased aerobic capacity also reduces anaerobic capacity

running more, without trying to imitate some image of perfect running form, is truly the best way to become a more skillful runner

If you do a long run at a steady pace, your intensity, by definition, will also remain steady, but your perceived effort will increase as fatigue sets in.

The fact that runners care about pace more than they do about perceived effort ratings and heart rates imbues it with the power to motivate in a way that other intensity metrics do not

Training cycles may be shorter than twenty-four weeks. The optimal duration is determined by your initial fitness level and by the distance of your peak race. If your peak race is long, such as a marathon, and your initial fitness level is fairly low, you should allow close to twenty-four weeks to prepare

Many elite runners do just one cross-training workout per week to supplement one hundred miles or more of running.

I read while I'm doing it. I get through a book a week this way

As a coach, I keep runners from turning easy runs into "somewhat hard" runs by giving them specific pace, heart rate, and perceived-effort targets for each workout

the experienced runner can run slow more often and get the same effect

Research has consistently shown that forced alterations to an individual's natural running form almost always worsen performance instead of improving it.

You will get the best possible results from your 80/20 training if you use all three of them complementarily

trained by heart rate in easy runs to keep herself from pushing too hard but trained by pace and time on the track.

Research and real-world practice have shown that the best way to achieve this objective is to sharply reduce the overall training volume while continuing to do high-intensity workouts. The contribution of low-intensity running to a well-executed taper may therefore be somewhat less than 80 percent.

If you're like many other runners, however, who tend to break down when they take chances on mileage, take heart: You can work around this limitation by supplementing your running with cycling or another form of nonimpact aerobic exercise

What's more, I find slideboarding much more fun than other indoor aerobic activities—almost as enjoyable as actual ice-skating.

The boundary between low intensity and moderate intensity, according to Seiler, falls at the ventilatory threshold, which is the intensity level at which the breathing rate abruptly deepens

during prolonged running at low intensity, the muscles release large amounts of a cell signaling compound called interleukin-6 (IL-6), which contributes to fatigue. Well-trained runners produce less IL-6, and this is one reason they are more fatigue resistant. It is believed that exposure to large amounts of IL-6 during running is itself the primary trigger of physiological adaptations that reduce IL-6 release in future workouts and boost endurance

brain fitness and aerobic capacity do not entirely account for improvements in running performance. As you will soon see, low-intensity, high-volume training also enhances performance in a way that is altogether separate from the two major components of running fitness.

If you want a better stride, all you have to do is run enough so that eventually you can run for two or three hours without looking any less fresh at the end than you did at the beginning and your form will “take care of itself.”

Zone 1 is appropriate for warm-ups, cool-downs, recovery periods between high-intensity intervals, and recovery runs

My guidelines for monitoring and controlling intensity by perceived effort, heart rate, and pace can be summarized in a few key points. Here they are:

- Start by doing either a thirty-minute time trial, a perceived effort-based test workout, or a talk-test workout to determine your lactate threshold heart rate, then calculate your five heart rate training zones using the percentages in Table 6.1.

- Use one of these same workouts to calibrate your perceived effort ratings, anchoring your lactate threshold intensity to a perceived effort of 6 on a 1 to 10 scale.
- Determine target pace ranges that correspond to your five heart rate zones either by finding the pace that corresponds with the heart rate marking the low end and high end of each zone or by using the McMillan calculator.
- Use perceived effort to establish your initial intensity whenever a workout calls for a change in intensity.
- Use heart rate as your primary intensity metric in all low-intensity runs (Zones 1 and 2).
- Use both heart rate and pace to regulate intensity when running at moderate intensity (Zone 3).
- “If you prefer not to train with a heart rate monitor, you may use pace to regulate your intensity whenever you would otherwise use heart rate, basing your target zones on Greg McMillan’s pace guidelines, but if you do this, it is imperative that you stay within those zones at all times, especially in low-intensity runs.
- Use pace as your primary intensity metric when running at high intensity (Zones 4 and 5), except when running uphill, in which case you should rely on perceived effort.
- Use perceived effort to fine-tune your intensity during your most challenging workouts. Specifically, use it to ensure you finish all such runs feeling you could have run at least a little faster or farther.
- Use perceived effort to adjust your intensity on “flat” days. Specifically, when you feel worse than normal at a given target heart rate or pace, slow down.

tempo run is a sustained effort in Zone 3 sandwiched between a warm-up and a cool-down. Tempo runs are a very effective means of enhancing a runner’s ability to sustain relatively aggressive speeds. They are emphasized in the peak phase of training for races of all distances from 5K to the marathon.

some elite runners still use pool running for cool-downs after long runs to pad their training volume without increasing injury risk.

Jeff Johnson, a brilliant mentor of young runners who had the distinction of being Nike's first employee and the man who named the company. Jeff's coaching philosophy was heavily influenced by that of Arthur Lydiard, a New Zealander who had revolutionized the sport in 1960 with a method that featured lots of slow, comfortable running and modest amounts of speed work

The key requirement for strengthening the aerobic system is stress.

inducing fatigue in the body through exercise triggers adaptations that make the body more resistant to fatigue in the future. Brain fatigue works the same way.

most of the work of coordinating the body's movements during locomotion was done not by the brain itself but rather by reflexes located throughout the body. The brain was involved only at an executive level—making decisions about which direction to turn and how fast to go and responding to novel circumstances that the reflexes were not prepared for

you can estimate your lactate threshold heart rate by starting a run at a perceived effort of 1 and then increasing your effort point by point until you give it a rating of 6.

THREE METRIC GUIDELINES FOR THE FIVE INTENSITY ZONES

long runs really need to be prescribed in distance to give every runner equal preparation to go the distance in their race.

It's okay to do more than one form of cross-training

you can actually race faster by training slower

The biggest performance gains were seen in the polarized training group

Fast runs may hurt more, but long runs hurt longer.

the brain never creates the same plan for the same action twice.

estimated time to exhaustion is also a very good subjective indicator of objective running intensity

Zone 3 efforts last long enough that cardiac lag is not an issue. Monitoring your heart rate during moderate-intensity runs will help you avoid pushing too hard, as it does in low-intensity runs. But moderate-intensity runs often are meant to be challenging, and monitoring your pace in them will motivate you to meet that challenge, as it does in high-intensity runs. So I suggest you use both heart rate and pace to regulate intensity in moderate-intensity runs, while using perceived effort to find the right intensity at the start of each Zone 3 effort (aiming for a rating of 5 to 6) and to make any necessary adjustments as you go, like slowing you down if you're having a flat day.

When selecting the workout types and levels to include in a given week, tally up the total time spent at low intensity and the total time spent at all intensities and divide the former by the latter to see how close the result comes to 0.80, or 80 percent. If it's not close enough, make an adjustment

It is not necessary, however, to follow the 80/20 Rule in running and cross-training individually. On the contrary, I believe it's best to do your hardest work in the discipline you're actually preparing to compete in. Therefore I advise you to do all of your cross-training workouts at low intensity and concentrate your full allotment of moderate- and high-intensity training within your runs.

Hard running isn't repeatable, either physically or psychologically. If you do too much of it, your body will burn out if your mind doesn't first.

I am trying to develop my runners until they are in a tireless state. In practice, this means I am trying to give them sufficient stamina to maintain their natural speed over whatever distance they are running. Stamina is the key to the whole thing, because you can take speed for granted.

At the elite level, runners who take up the sport in high school typically attain their lifetime peak VO₂max by their second year of college. Their best race times, however, often don't come until a full decade later.

each runner's stride automatically becomes more efficient over time so that conscious changes in technique are always unnecessary and usually counterproductive

you must always allow perceived effort to have the final say in controlling the intensity of workouts. Even though perceived effort is “all in your head,” it tells you how your body is doing in a holistic way that pace and heart rate do not.

The trick is to push yourself without pushing too hard in these faster runs

According to the World Health Organization, daily aerobic exercise is required for maximum all-around health

it's sensible for experienced runners to transition to doing more cross-training and less running after age thirty-five. Those who do so stand to extend their peak performance years and ultimately reduce the rate at which their performance declines

80/20 Rule. The five-zone intensity scheme that you use for running may also be applied to your swimming and cycling. However, because the lactate threshold heart rate is different for each activity, you will need to establish distinct training zones for swimming, cycling, and running. You can determine you

SAMPLE “WEEK OF SLOW” SCHEDULES

A high aerobic capacity is not intrinsically beneficial. No prizes are awarded for VO₂max numbers

all great runners exhibit a subtler quality that we might call relaxed smooth ease.

perceived effort is literally “all in your head.”

To avoid getting tripped up by cardiac lag in workouts involving short efforts at high intensity, use perceived effort and pace as your primary intensity metrics in such workouts. Changes in perceived effort are instantaneous, and most running devices with GPS or accelerometer technology register changes in pace within a few seconds

if you're training for a marathon, you should probably do more moderate-intensity running because it is more specific to the actual demands of marathon racing.

I believe that most runners, including those who seldom get injured, should do at least one nonimpact cardio workout per week when healthy. The reason is that runners who make cross-training a part of their normal routine are more prepared physically and psychologically to fall back on an alternative form of exercise in times of crisis

Slideboarding strengthens the muscles on the inside and outside of the hips that tend to be weak in run

My role is not to innovate and discover, like Arthur Lydiard and Stephen Seiler, but to serve as a link between the innovators and discoverers and the broader running community.

the effects of low-intensity and high-intensity exercise on the aerobic system are somewhat complementary. Maximizing aerobic capacity therefore requires that a runner do some of both

To be a skillful runner is to have an efficient stride. The standard measure of stride efficiency is running economy

intensity, which we can define simply as how hard the body is working during a run relative to how hard it is capable of working

Zone 4 is typically targeted in high-intensity interval workouts featuring longer intervals (two to eight minutes) and in fartlek (or what I call “speed-play”) runs, which are simply interval workouts with a less formal structure that are done on roads or trails instead of at the track

Do approximately 80 percent of your training at low intensity and 20 percent at moderate and high intensities, except during specific periods of training when it is beneficial to do somewhat more or somewhat less low-intensity training, and unless you learn through experience that you benefit from slightly greater or slightly lesser amounts of low-intensity running, and be sure to vary the balance of moderate- and high-intensity training in your program based on the sort of race you’re preparing for.

Before you begin the plan, build your training to the point where you are running at least three times per week, including some short efforts at moderate and high intensities and some easy runs of at least eight miles, and where you are doing aerobic exercise at least seven times per week

it’s best to mix pool running with a weight-bearing activity such as elliptical training during extended periods of injury rehabilitation

Seiler to embark on a research agenda that culminated in the most significant breakthrough in running since Arthur Lydiard's original discovery of slow running: the 80/20 Rule.

increasing the intensity beyond that threshold brings additional benefits.

the duration of exercise matters far more than does the intensity of exercise with respect to the goal of enhancing fatigue resistance in the brain

Becoming a more skillful and efficient runner is more like growing a beard than it is like chopping wood. In other words, you don't make it happen—you allow it to happen.

5-ZONE HEART RATE TRAINING SYSTEM

If you use Greg McMillan's pace guidelines, recalculate your pace zones after each race, using your latest finish time as the new input.

Fast finish runs, tempo runs, long runs with speed play, and long runs with fast finish include efforts in Zone 3.

If you intend to do a lot of cycling, it's important that you use a bike that fits you properly. A proper bike fit is much more complex than a proper running shoe fit

the secret to becoming a speedier runner is going slow most of the time.

athletes generally get more out of time spent at low intensity than they get out of time spent at moderate intensity

Mostly-slow training also does a better job of strengthening the parts of the brain where tolerance for suffering resides.

The harder the brain is working to control movements, the more constrained those movements become. Better athletes don't have to exert as much mental effort to control their sport-specific movements, so their movements are freer

effective heart rate-based training requires that target heart rate zones be individually customized.

Be sure to conduct this workout on flat terrain. Begin by jogging very slowly for five minutes. Next, adjust your effort until your heart rate settles into a steady rhythm at the top of your Zone 1 heart rate range. Note your pace. Now increase your effort very slightly until your heart rate is at the bottom of your Zone 2 range. (This will be no more than two BPM higher.) Again, note your pace. Continue in this fashion until your heart rate reaches the bottom end of your Zone 5 range and note the corresponding pace one last time. You now have a full set of pace zones to complement your heart rate zones.

A recovery run is a run undertaken entirely in Zone 1. These runs are intended to be done as the next run after a very challenging workout

Activities that involve alternating movements of the legs, such as cycling, work very well. Upper-body dominant activities such swimming do not.

a runner gets faster by going slow in the majority of his or her runs.

Runners routinely measure and track their volume, but they do not monitor their intensity distribution in the rigorous manner that scientists have begun to do

a brain-based tolerance for suffering plays as big a role as aerobic capacity does in relation to fatigue resistance.

cyclists naturally adopt a pedaling cadence that does not quite minimize oxygen consumption but does minimize muscle activation

trust your brain more than you trust your GPS watch or your heart rate monitor and slow down as much as necessary to keep your perceived effort where it should be.

pace is free of the lag phenomenon that limits the usefulness of heart rate monitoring in high-intensity workouts featuring short intervals in Zone 4 or 5

I would like to persuade you to want to run more than you do today

A sensible upper limit for total workouts in a week (running plus cross-training) is thirteen—that's two workouts per day on six days of the week plus one workout on a designated recovery day. Of these thirteen workouts, no more than six or seven need to be runs, even if you're 100 percent committed to becoming the best runner you can be

The process of getting in shape for races should be a lopsided effort to increase stamina to the point where a runner could run more or less forever. The secret to running faster was to run farther.

developing the aerobic system reduces top-end speed.

relaxed smooth ease is the visible manifestation of an acquired “quietness” in parts of the brain that control the body's movements, a sort of neurological efficiency earned through relentless practice that allows the stride to be more adaptive. The ultimate mark of skillful running is the ability to run with minimal mental effort.

The harder the brain is driving the muscles, the more intense the effort feels.

runners seldom try as hard when running entirely by feel

The human body is not able to absorb increasing training loads for longer than about twenty-four weeks at a stretch. For this reason, runners need to train in cycles. Each training cycle must be followed by a recuperative period lasting at least a couple of weeks. Despite losing some fitness during such breaks, you will be able to start the next cycle at a higher level of base fitness than you did the last one, and therefore you will also be able to train harder and attain a higher level of peak fitness.

But my goal here is to encourage you to use cross-training for the additional purpose of increasing your overall training volume and enhancing your fitness. To fulfill this purpose, you must incorporate cross-training into your program in a manner that increases your overall training volume.

When you first try uphill treadmill walking, you will have to play around with different speeds and gradients to find those that place you in Zones 1 and 2. Be sure to keep your first few sessions relatively short to avoid straining your calf muscles and Achilles tendons. After a couple of weeks, these tissues will be fully adapted to the activity.

PEAR Mobile app, which uses my voice to guide runners through heart rate-based workouts, and developed a separate 80/20 Running app that keeps track of time spent at low and moderate to high intensities.

IL-6 release is affected by the duration of running more than the speed

The elements of correct running technique, according to tradition, include a high stride rate, a low stride length (relative to speed), a midfoot or forefoot foot strike, and minimal vertical oscillation (or bouncing)

The three most practical measures of running intensity are perceived effort, heart rate, and pace.

Note that your lactate threshold heart rate will change somewhat as your fitness level changes, rising slightly as you get fitter and dropping a bit if you should ever lose fitness

The taper period immediately precedes an important race

where you are running at least three times per week, including some short efforts at moderate and high intensities and some easy runs of at least ten miles and where you are doing aerobic exercise at least seven times per week.

It provides an excellent aerobic-training stimulus through a weight-bearing action involving alternating use of the legs, just like running

Seiler learned that elite cyclists, swimmers, triathletes, rowers, and—yes—runners did the same thing

However, there are some things that low-intensity exercise does better.

long runs at low intensity represent the most effective way to boost the brain's capacity to resist fatigue

Given what we now know about the self-optimizing nature of the stride, we should indeed expect that simply running more would be the surest way to accelerate the acquisition of relaxed smooth ease.

Zone 3 is targeted in tempo runs, cruise intervals, and fast finish runs. A typical threshold run comprises a steady effort of fifteen to forty minutes in Zone 3 sandwiched between a warm-up and a cool-down. Cruise intervals are like tempo runs except the middle of the sandwich is separated into multiple shorter Zone 3 efforts. A fast finish run is a foundation run with a short (five to fifteen minutes) effort in Zone 3 tacked onto the end

Due to cardiac lag, however, heart rate monitoring itself is not helpful when you are running short uphill intervals at high intensity. In this case, it's best to use perceived effort as your primary intensity metric while also paying attention to your time to complete each hill interval. Aim to match or slightly improve those times when you repeat the workout.

way to infuse a little moderate-intensity running into your training during the base phase, when your focus is on increasing volume, and during recovery weeks

ElliptiGO bikes are made with racing-quality materials and parts, so they aren't cheap: \$2,200 and up.

runners who strictly limit their faster running in workouts derive more benefit from these sessions and perform better in races

The human aerobic system includes the heart and lungs, the arteries and capillaries that deliver blood to the muscles, the blood itself, and mitochondria, which are little “factories” within muscle cells where oxygen is used to release energy from metabolic fuels.

prolonged exercise at low intensity fatigues the brain to a much greater degree than does high-intensity exercise that doesn’t last as long

The first walking robots could do only what they were programmed to do in exactly the way they were programmed to do it. They could not learn to do it better or alter their movements in response to changes in their environment (such as encountering a slope).

thirty-minute time trial. Here’s how it works: Warm up with several minutes of easy jogging and then run as far as you can in thirty minutes while wearing your heart rate monitor. Your average heart rate during the last ten minutes of this effort is your lactate threshold heart rate. The downside of this test is that it’s rather painful.

McMillan’s pace guidelines for high-intensity intervals are all distance-based, whereas some coaches and training plans use time-based workouts. Most of the workouts you will see in the next chapter, in fact, are time based. To perform time-based high-intensity workouts with Greg McMillan’s pace guidelines, you will need to convert his suggested times for intervals of specific distances into a pace in minutes and seconds per mile.

foundation run is a steady, low-intensity run of short to moderate duration. It begins with a warm-up in Zone 1, then moves to Zone 2 for a while, and concludes with a cool-down back in Zone 1. Foundation runs are the bread and butter of effective 80/20 training—the workout type you’ll rely on more than any other to ensure you’re doing enough running at low intensity.

If, for example, you live in an urban area where cycling is a hassle but you happen to own a treadmill, then uphill treadmill walking might be your best option

Do 80 percent of your running at low intensity and the other 20 percent at moderate to high intensity

When they run in groups,” he told me, “sometimes some of them go at a little higher intensity than the prescribed one.” Any runner who has experienced group training is familiar with this effect. The strongest runner sets the pace, and the others follow like lemmings.

Low-intensity, high-volume training develops the sort of suffering tolerance that enhances fatigue resistance more effectively than does speed-based train

Variation in the running stride is best thought of as a kind of freedom of movement, or looseness

TEN-POINT SCALE OF PERCEIVED EFFORT

In low-intensity workouts (Zones 1 and 2), pace should not be your primary intensity metric

Besides the matter of willingness, a separate impediment to running more is fear of injury. In chapter 12, I will show you how to work around this issue by using cross-training as an alternative to running more.

Regardless of how you choose to balance running and cross-training, it's essential that you apply the 80/20 Rule to your total training regimen, not just your running

in order to yield steady improvement, a training system must be repeatable

Lydiard discovered that he felt strongest when he ran one hundred to 120 miles per week and varied the distance of the individual workouts from day to day

Arthur Lydiard argued that distance runners do not need to increase their maximum speed. He was right about that. But it turns out that runners can't increase their maximum speed if they train for increased aerobic capacity

the stride is a self-optimizing system

use perceived effort to establish the appropriate intensity at the start of a workout or workout segment. Then rely on pace or heart rate to maintain that intensity while allowing your perceived effort to gradually increase as fatigue sets in

Seldom, if ever, should you try as hard in workouts as you do in races. Treating workouts like races is a recipe for burnout.

Once you have corrected the intensity balance in your training, consider increasing your running volume

schedule of running once a day in the morning and doing a nonimpact aerobic workout in the afternoon

Use the following formula to calculate your critical velocity, which is also your lactate threshold swim pace. Critical velocity = $(400 \text{ yards} - 200 \text{ yards}) \div (400 \text{ time} - 200 \text{ time})$

Heart rate monitoring in particular is an effective tool for getting runners to slow down, while pace targets are better for getting runners to push themselves in the 20 percent of their workouts when they're supposed to

larger amounts of faster running offer no additional benefit

Forget about form," he wrote. "If a joker throws his arms around, that's fine, so long as he is fit and relaxed. Then he runs smoother and easier, and form takes care of itself. We want the chap who can run for two or three hours and come back looking as fit as he did when he went out.

Perceived effort is nothing more than how hard you feel you're working during a run.

Heart rate monitoring is less useful in high-intensity runs

some intriguing studies have demonstrated that endurance athletes get great results from a “polarized” training approach where they do 80 percent of their training at low intensity, 20 percent at high intensity, and none at moderate intensity. On the other hand, the most impressive results obtained so far in any study of the effects of 80/20 training on runners were seen in runners who did roughly equal amounts of moderate- and high-intensity training. There is also evidence that athletes get more out of moderate-intensity training than they do out of high-intensity training when preparing for longer events.

One benefit of running that cycling and other alternatives can’t match is improvement in running skill

Despite its lack of familiarity, slideboarding is a great cross-training choice for runners who prize convenience and affordability and are susceptible to knee pain.

In well-trained runners, the ventilatory threshold typically falls between 77 percent and 79 percent of maximum heart rate

the primary trigger for IL-6 release is depletion of glycogen fuel in the working muscles. Long, slow runs cause much higher levels of glycogen depletion

fitness is not the only factor that determines running performance. A separate factor is running skill.

Another simple do-it-yourself method of determining the lactate threshold heart rate is the talk test. The lactate threshold heart rate is associated with the fastest running pace at which a runner is able to talk comfortably. While wearing a heart rate monitor, start jogging at a slow, steady pace. After one minute, recite the Pledge of Allegiance (see Figure 6.3) and note your heart rate. Alternatively, count upward from 120 to 125. Now increase your speed slightly, wait a minute, recite the Pledge (or count) again, and again note your heart rate. Continue in this fashion until talking is slightly uncomfortable. Note your pace at the previous speed when speaking was comfortable or sort of comfortable. Your heart rate at that pace is your current lactate threshold heart rate

Zone 2 is targeted in what I call foundation runs (which most other coaches call easy runs) and long runs

Recovery runs, foundation runs, and long runs include efforts in Zones 1 and 2 only.

the half marathon requires well-rounded running fitness

When running in water, you should make your best effort to mimic a natural overground running stride as closely as possible. This is next to impossible if you're not wearing a pool running vest such as those made by AquaJogger,

Stephen Seiler, an American exercise physiologist based in Norway

you need to elevate your heart rate to at least 60 percent of maximum

Shorter, faster runs aren't as effective at fatiguing and strengthening these brain areas, which include the insular and temporal lobes, whose job is to perceive the physical elements of emotional states (such as the discomfort of extreme effort), and the anterior cingulate cortex (ACC), whose job is to resolve internal conflicts (such as the conflict between the desire to keep going and the desire to quit when one is running in a fatigued state)

the stride pattern that is most efficient for you when you're fresh is no longer optimal when you're tired.

If your breathing returns to its pretalking rhythm by the third breath after you finish speaking, then you were able to talk comfortably. Note that the smaller your speed increases are when you perform this test, the more precise the result will be.

SUMMARY OF INTENSITY MONITORING METRICS

long run is simply an extended foundation run that is measured in distance instead of time

Runners who use these machines for supplemental training usually run at an effective body weight of 90 to 95 percent.

four-fifths of their total training time below the ventilatory threshold (VT), or running slow enough to carry on a conversation

high-intensity training is essential, but that a little goes a long way.

low-intensity, high-volume training teaches the mind to accept that it might as well make peace with its suffering because it won't end anytime soon.

Experienced and skilled athletes in all sports exhibit less brain activity, or what I call a quieter brain

if you want to increase your muscles' ability to burn fat, all you have to do is strap on a heart rate monitor and run at your ventilatory threshold heart rate.

Once you've established your heart rate zones, complete another test workout to establish matching pace zones. To do this test you will need a device such as the Polar RS300X that is capable of measuring both heart rate and pace.

Note that in the case of interval-type workouts that feature relatively short active recovery periods between high-intensity efforts, the entire interval section of the workout, including recovery periods, is counted as time spent at high intensity.

Exempt yourself from this advice, however, if you are prone to suffering injuries such as Achilles tendon strains and hamstring strains that are associated with faster running. In this case, concentrate your full allotment of moderate- and high-intensity training within your chosen cross-training discipline.

There is an obsession these days with high intensity. Most of the trendy new training systems are focused on speed work

TYPICAL WEEK OF TRAINING FOR AN ELITE KENYAN RUNNER

The traditional VO₂max test ties together oxygen consumption and fatigue in an artificial way that gives the appearance that aerobic capacity is the one and only factor limiting endurance performance

the brain in fact monitors muscle activation and latches onto patterns that yield equal speed with less muscle work—a process that would reduce the oxygen cost of running as a side effect.

Every runner has occasional flat days in training. Often these days come when you least expect them and have no apparent cause. But there's always a cause, whether it's identifiable or not, and this feeling of flatness is your brain's way of telling you that it has detected something amiss internally and you shouldn't push yourself. On these days, you will experience unusually high effort at any given pace or heart rate. If you force yourself to stick to a preselected pace or heart rate target, you will probably end up feeling even worse the next day

aim to slightly improve your performance in benchmark workouts as the training process unfolds, instead of aiming to achieve your best possible performance in each of them as you would do in a race

Not only will your health improve but your running will too

The more concerned you are with maximizing performance, the more cross-training workouts you should do.

u use for runnin

A typical week, after he'd gotten it all figured out, looked something like this: Monday: 10 miles Tuesday: 15 miles Wednesday: 12 miles Thursday: 18 miles Friday: 10 miles Saturday: 15 miles Sunday: 24 miles

a higher aerobic capacity will not do for you, and that is make you faster in any pure sense

The more relaxed a runner's stride is, the more resistant that runner is to fatigue.

effort perception is mainly a function of how hard the brain is working to drive the muscles

This makes pace a reliable indicator of intensity, because there is a direct relationship between how much the body puts into running and the resulting performance output. If you're running faster, it can only be because your body is working harder.

Assuming you proceeded in a sensible way, increasing your training load very slowly and giving your body regular opportunities to recover, you would make consistent progress for up to twenty-four weeks. Then you would hit a wall. At that point, any further increases in

your training load would yield no further gains in fitness and instead would only leave you more and more fatigued

If you immediately go from running every day to, say, cycling every day because of any injury, there's a good chance you will also suffer a cycling-related overuse injury.

Uphill treadmill walking is better than regular walking in two ways. First, while it's hard to get the heart rate up into the Zone 2 range on an outdoor walk in most environments, with uphill treadmill walking it's easy

unless a runner is systematically held back, he will more often than not run too hard on easy days and unwittingly sabotage his training plan.

Research shows that low-intensity training is truly a gift that keeps on giving. The more slow running you do—up to a point, of course—the more you get in return

running economy is represented as the amount of oxygen required to sustain a given speed

you need to actively monitor and control your intensity throughout every run

Zone 5 is appropriate for high-intensity runs featuring shorter intervals (thirty to ninety seconds), hill repetition runs, and more intense speed-play runs.

The training process starts with a base phase. The primary objective of this phase is to gradually increase running mileage to a target level that is appropriate for the individual runner. In this case it is generally best to do more than 80 percent of training at low intensity, because it's both easier and less risky to increase running mileage when faster running is kept to a modicum. It is important to include some faster running in the base phase, though, because it prepares the body for the harder workouts of the peak phase, when the 80/20 Rule goes into effect.

marathon distance is, in my opinion, a bit longer than humans were meant to race on foot

Slideboarding is typically done by hockey players and speed skaters when they can't get on the ice to train.

Seiler started by exhaustively analyzing the training methods of world-class rowers and cross-country skiers. He found a remarkable consistency: Athletes in both sports did approximately 80 percent of their training sessions at low intensity and 20 percent at high intensity.

high-intensity exercise strengthens the aerobic system faster than does low-intensity exercise

brain fitness makes a major contribution to physical endurance that is completely independent of physiology below the neck. You don't even have to exercise to increase brain fitness and fatigue resistance in endurance exercise. All it requires is sustained mental focus on a cognitively demanding task

reduce brain activity during the act of running. Any measure that helps a runner run more unconsciously is guaranteed to result in better performance

Zones 1 and 2, then, represent low intensity and zones 4 and 5 are high intensity.

Heart rate is a more useful metric for monitoring and controlling intensity on hills because a heart rate of, say, 136 beats per minute represents the same physiological intensity whether you're running uphill, downhill, or on level ground

fast finish run tacks a relatively short effort in Zone 3 onto the end of a foundation run.

you need to make sure your saddle height, saddle fore-aft position, crank arm length, handlebar position, and several other settings are dialed in just right. This requires that you get a professional fitting from a certified bike fitter

Running too hard too often is the single most common and detrimental mistake in the sport.

The physiological cornerstone of running performance is aerobic capacity, or the body's ability to extract oxygen from the environment and use it to release energy from muscle fuels

Runners often observe that very long runs and races are more mentally challenging than shorter ones

In runners, a quiet brain yields a subtle play in the stride that observers may perceive as a sort of effortlessness.

In my five-zone scheme, the lactate threshold heart rate marks the upper limit of the moderate intensity range. Heart rates above this number represent high intensity. Heart rates below the ventilatory threshold (which, again, is just a step below the lactate threshold) represent low intensity.

There are many pace-based training systems that prescribe individualized target pace zones for runners based on some measure of their current running ability, such as a recent race time. My favorite among them is the one developed by Greg McMillan, which you can find on his Web site at mcmillanrunning.com. To use it, enter a recent race time or a goal time for a race of any distance between 800 meters and a marathon